

1 Fig. 1.1, 1.2 and 1.3 are electron micrographs taken from different cells.

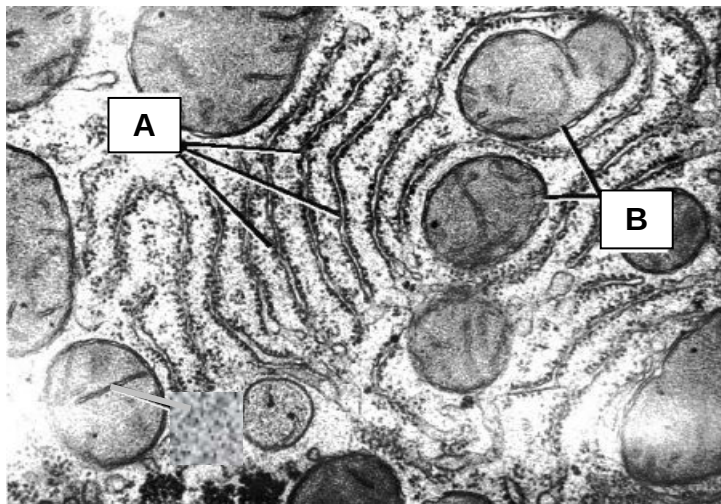


Fig 1.1

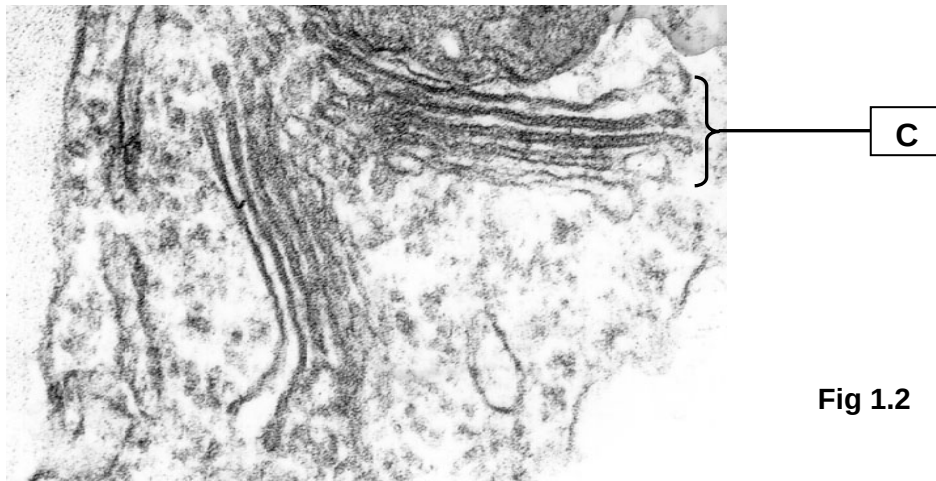


Fig 1.2

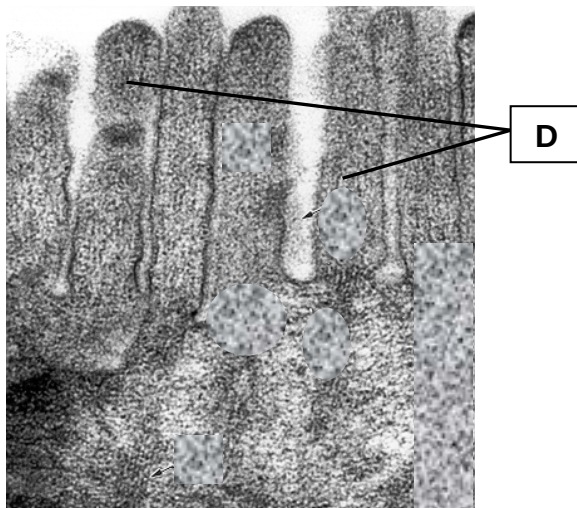


Fig 1.3

(a) Name the structures labeled A to D.

A RER/ ribosomes

C Golgi body/apparatus

B Mitochondria

D Microvilli

[2]

@ ½ mk

(b) Describe briefly how the two structures, **A** and **C**, are related in the cell.

Polypeptides synthesized at ribosomes are channeled into RER;

where post-translational modifications/ref. to glycosylation take place*;

Intracellular transport via interconnected cisternae;

and transport vesicles to Golgi apparatus;

where final products (eg ref. to lysozyme) are formed and packaged (into secretory vesicles/lysosomes) ; @ 1mk

[3]

(c) With reference to Fig. 1.1, explain the significance of the numerous **B** found adjacent to **A**.

Synthesis of ATP;

required for protein synthesis/translation at ribosome / RER; @ ½ mk

[1]

(d) (i) State the function of structure **D**.

Increase surface area of cell surface membrane;

Ref. to uptake of material/nutrients; @ ½ mk

[1]

(ii) Suggest where such structures may be found in the human body.

Small intestine/ kidney / tastebuds / inner ear / smell receptors; @ 1mk

[1]

[Total: 8]

- 2 In an experiment to study the effect of heat treatment on the digestibility of protein substrate and the effect of raw bean extract on protease activity, various reaction mixtures were prepared and were incubated for 30 minutes.

The protein concentration of each reaction mixture at the beginning and at the end of the incubation period was determined by the colorimetric method which measures colour intensity of these reaction mixtures. The results were shown in table 2.1.

Table 2.1

Incubation period /min	Colour intensity of the reaction mixture / arbitrary unit		
	Tube A	Tube B	Tube C
	Protease + heated protein substrate	Protease + unheated protein substrate	Protease + unheated protein substrate + raw bean extract
0	10	10	10
30	4	7	9

Fig. 2.1 shows a standard graph obtained by using colorimetric method for determining concentration of protein solutions.

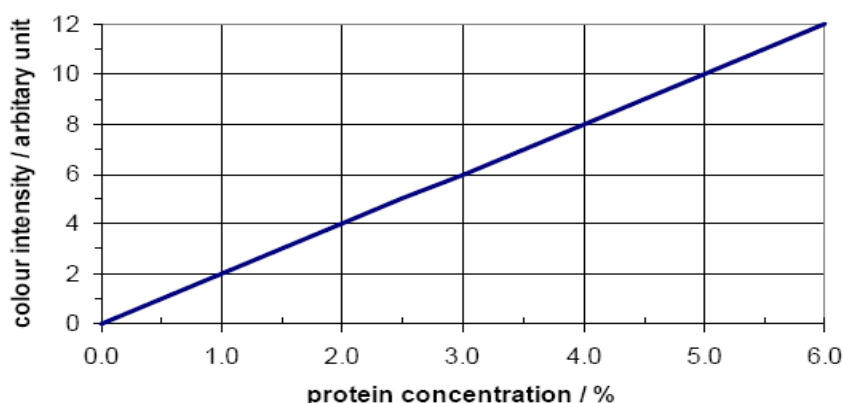


Fig 2.1

(a) With reference to Table 2.1 and Fig 2.1,

- (i) Determine the protein concentration of the three tubes at time 0 min.

5% protein concentration; @1 mk [1]

- (ii) Explain which protein substrate is more digestible by protease.

heated protein more digestible by protease;

less protein remained after incubation with protease / lowest colour intensity;

unheated = 3.5% protein concentration;

vs heated protein = 2% @½ mk [2]

- (b)** Based on your knowledge of protein structure, explain the effect of heat treatment in the digestibility of this protein substrate.

heating increases the digestibility of this protein substrate; (award this even if wrong concept to explain)

heating causes protein substrate denaturation/ loss of 3D configuration;

breaking of bonds – H, hydrophobic, ionic, disulphide (name at least 2);

increase in surface area of substrate for protease action;

Ref. to easier access of enzymes' active site to substrate; @ 1mk [3]

- (c)** Suggest the effect of raw bean extract on protease activity.

inhibits protease activity/competitive/non-competitive inhibitor;

***R!* inhibitor to substrate**

@ 1 mk [1]

- (d)** Fig 2.2 shows the structure of a biochemical molecule, **W**.

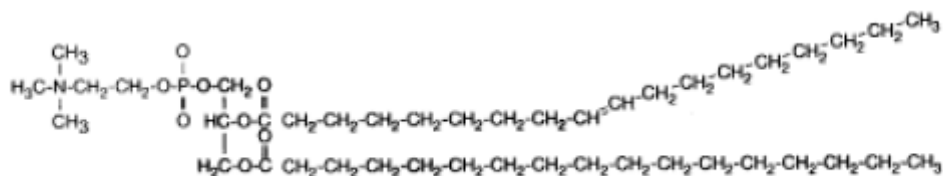


Fig 2.2

- (i) Identify the molecule **W**.

phospholipid; @1mk [1]

- (ii) Proteins can associate with this molecule in the cell membrane. Based on your knowledge on the structure of amino acids/proteins, give a reason for this observation.

proteins/amino acids have both hydrophobic;

and hydrophilic side chains/R-groups:

can bind to corresponding components of molecule W;

polar, hydrophilic R groups can form ionic bonds/ H bonds,

while non-polar/hydrophobic R group can form hydrophobic interactions:

@ ½ mk [2]

[Total:10]

- 3 (a) Explain how the promoter of a eukaryotic gene influences transcription.

It is a (regulatory) non-coding (DNA) sequence (found at 5' end of a gene);
allows binding of general transcription factors to TATA box;
allows binding of RNA polymerase (II) ;

to form transcription initiation complex / allows for initiation of transcription;

Ref to. promoter works in concert with other (proximal/distal) control elements to control the level of transcription; @ 1 mk [3]

- (b) Fig. 3.1 shows the binding of a general transcription factor to part of the promoter. This will help to recruit RNA polymerase to the promoter.

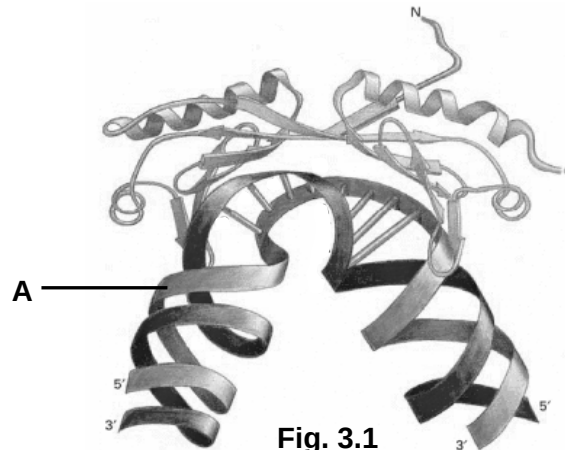


Fig. 3.1

- (i) With reference to Fig. 3.1, give **two** reasons why the structure labelled **A** is part of the promoter and not the transcription factor.

Promoter is made of DNA;
Ref. to double helix / stranded;

Has 5' and 3' ends;

which are antiparallel; @ 1mk [2]

- (ii) Suggest why a substitution mutation at the promoter may result in the silencing of the gene controlled by the promoter.

Ref. to new sequence not complementary in shape to binding site of / not recognised by transcription factors / RNA polymerase;

Transcription factors / RNA polymerase cannot bind;

Transcription cannot be initiated; @ 1mk [2]

- (c) Control of gene expression occurs mostly at the level of transcription in both eukaryotes and prokaryotes. State **two** ways in which control of gene expression in eukaryotes and prokaryotes are different.

Genes grouped into operons, all genes of an operon are expressed simultaneously vs each gene has its own control elements;

Control at the level of post-transcription not available vs available;

Control at level of translation not prominent vs prominent;

@ 1mk [2]

[Total: 9]

- 4 Scallops, which are bivalve mollusks, are important commercially throughout the world. The marine bay scallop, *Agropecten irradians*, has three distinct shell colours, yellow, orange and black. The shell colour is controlled by a gene with three alleles, yellow (S^y), orange (S^o) and black (S^b).

Scallops are hermaphrodite and are able to fertilise themselves to produce offspring.

Single mature adult specimens of yellow, orange and black scallops were collected and kept in the separate tanks of seawater until they produced young. The young were then scored for shell colour. The results were as follows:

Parent	Offspring
Yellow scallop	25 yellow and 8 black
Orange scallop	31 orange and 9 black
Black scallop	27 black

- (a) (i) Using genetic diagrams and the symbols given, explain the results from the orange and black scallops. [5]

Orange scallop

Parent	$S^o S^b$	x	$S^o S^b$	;
Gametes	(S^o)	(S^b)	(S^o)	(S^b) ;
F1 genotype	$S^o S^o$	$S^o S^b$	$S^o S^b$	$S^b S^b$;
F1 phenotype	Orange	Orange	Orange	Black ;
F1 ratio	3 Orange: 1 Black			;

Black scallop

Parent	$S^b S^b$	x	$S^b S^b$	;
Gametes	(S^b)		(S^b)	;
F1 genotype	$S^b S^b$			;
F1 phenotype	(All) Black			::

- (ii) Orange scallops are more valued for human consumption.

Describe how a marine biologist could produce a pure-breeding line of orange scallops for commercial exploitation using the offspring from the single orange scallop from (i).

Cross orange scallops produced from first cross with black scallops OR

Let orange scallops self-fertilise;

Select those that produce only orange offspring;

@ 1 mk

[2]

[Total 7]

- 5 The polymerase chain reaction (PCR) is a method of obtaining large amounts of DNA from a small initial sample. Fig. 5.1 shows the main stages in the polymerase chain reaction.

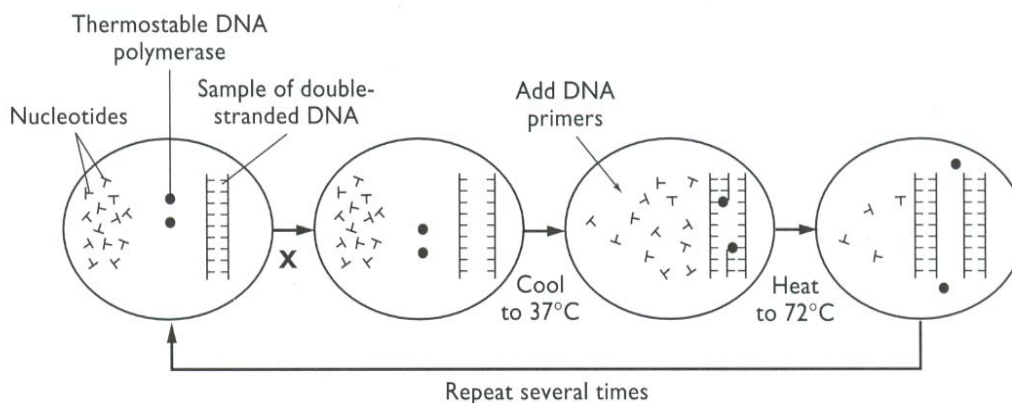


Fig. 5.1

- (a) (i) Explain why DNA synthesis takes place in opposite directions on the two separated strands of the double helix.

DNA strands are antiparallel;

Taq/ DNA polymerase only adds nucleotides to the free 3' end of a growing DNA strand / synthesizes new strands in the 5' – 3' direction;

@ 1mk

[2]

- (ii) A primer is a small piece of synthetically manufactured single-stranded DNA with a specific nucleotide sequence. State why the primer's specific nucleotide sequence is critical to its function in the PCR.

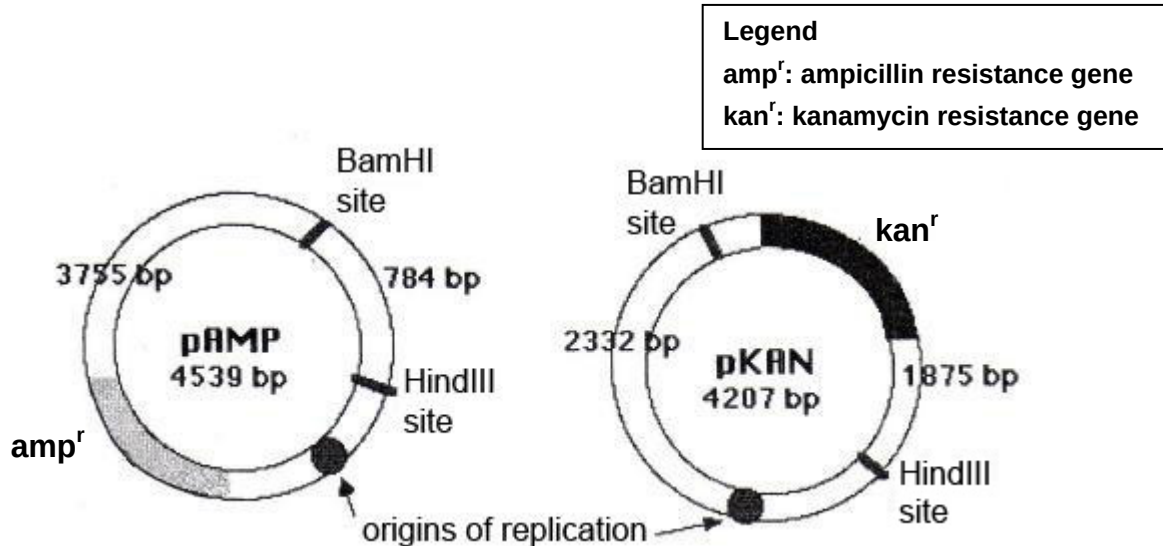
Primer has to be complementary to a specific sequence of DNA at ends;

Ref. to flanking/specifying the region of interest to be amplified; @ ½ mk

[1]

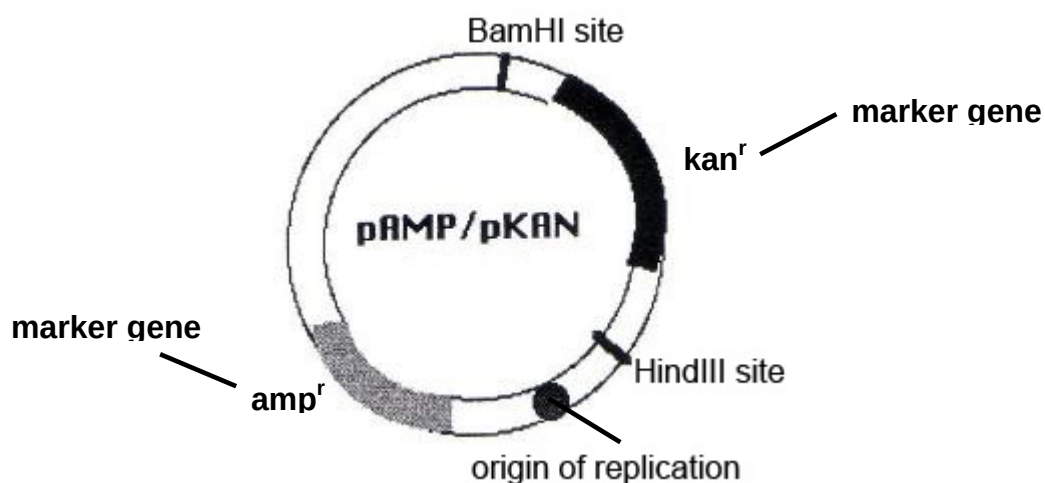
In the process of cloning a eukaryotic gene, bacterial cells that contain the gene of interest can be identified by a double selection process using two antibiotic resistance markers. The cloning vector can be made from two plasmids (pAMP and pKAN) shown below.

Both pAMP and pKAN contain one *Bam*HI site, one *Hind*III site and one origin of replication each. pAMP contains the ampicillin resistance gene while pKAN contains the kanamycin resistance gene.



To form the desired cloning vector, both pAMP and pKAN are treated *Bam*HI and *Hind*III. The resulting four restriction fragments are allowed to ligate to form several possibilities of circular plasmids, among which, only one of them is the desired outcome which consists of two antibiotic resistance genes.

- (b) (i) Draw and annotate the desired cloning vector formed from pAMP and pKAN. Annotate with the origin of replication, both marker genes, and the *Bam*HI and *Hind*III sites.

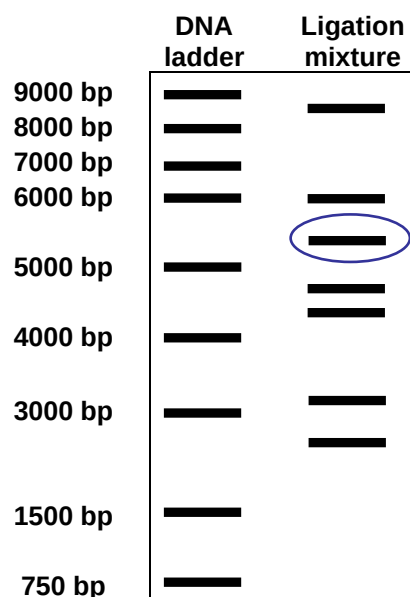


Drawing showing a cloning vector with 2 resistance genes and origin of replication;
 Labelling origin of replication, kan^r, amp^r, BamHI site, HindIII site ;;;
 - ½ mk for one mistake, -1 for two mistakes, -1½ for three mistakes

[2]

- (ii) Gel electrophoresis was performed to separate the ligation mixture which contains several circular plasmids. A DNA ladder consisting of markers of known sizes was also loaded onto the agarose gel.

Circle the band which indicates the position of the desired cloning vector.



[1]

[Total: 6]

Section B

Answer only **one** question.

Write your answers in the lined pages provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 6 (a) Explain the effects of pH on the rate of enzyme-catalysed reactions. [5]
- (b) Describe the main structural features of cellulose and collagen. [8]
- (c) Describe how carbohydrates are formed in the Calvin cycle. [7]
- 7 (a) Explain the unique features of stem cells and, using appropriate named examples, describe briefly the normal functions of these cells in the body. [6]
- (b) What are the benefits that can be derived from sequencing the human genome? [7]
- (c) Discuss briefly the ethical issues raised by the human genome project. [7]

6 (a) Explain the effects of pH on the rate of enzyme-catalysed reactions. [5]

- 1 Enzymes work best at an optimum pH;
- 2 When pH changes from optimum, rate of reaction drops;
- 3 ionic charges of acidic and basic groups of residues reassociate;
- 4 Binding residues are unable to bind to substrate / ES complexes cannot form;
- 5 Catalytic residues are unable to catalyse reaction;
- 6 (At extreme pH changes,) ionic bonds / H bonds are disrupted;
- 7 Protein/active site loses specific 3D shape / denatured;

@ 1mk max 5

(b) Describe the main structural features of cellulose and collagen. [8]

Cellulose

- 1 a polysaccharide;
- 2 of β glucose;
- 3 linked by 1, 4;
- 4 glycosidic bonds;
- 5 alternate glucose units are inverted 180° ;
- 6 linear;
- 7 OH groups alternate of both sides of the chain;
- 8 cross-links between adjacent cellulose chains;
- 9 ref. to microfibrils;
- 10 ref. to bundles of microfibrils/ macrofibrils/ fibres;

Collagen

- 11 a fibrous protein;
- 12 made of amino acids;
- 13 ref. to peptide bonds;
- 14 repeating tripeptide sequence of Gly-X-Y (where X is often proline and Y is often hydroxyproline);
- 15 two novel amino acids, hydroxyproline and hydroxylysine;
- 16 ref. to each chain helical;
- 17 triple helix/ tropocollagen;
- 18 Held together tightly by hydrogen bonds between chains;
- 19 ref. to staggered ends;
- 20 chains are cross-linked covalently;
- 21 bundled to form collagen fibres;

@ $\frac{1}{2}$ mk each max 8

(c) Describe how carbohydrates are formed in the Calvin cycle. [7]

- 1 CO_2 combines with 5C sugar ribulose bisphosphate (RuBP);
- 2 Catalysed by RuBP carboxylase/Rubisco;
- 3 6C intermediate splits to form two 3C compound;
- 4 glycerate-3-phosphate (GP);
- 5 use of ATP;
- 6 reduction / use of NADPH;
- 7 to form 1 glyceraldehyde-3-phosphate (GALP/triose phosphate);
- 8 GALP/G3P used to regenerate RuBP,
- 9 utilising ATP molecules;
- 10 GALP/G3P/triose phosphate used for synthesis of carbohydrate;

@ 1 mk max 7

- 7 (a) Explain the unique features of stem cells and, using appropriate named examples, describe briefly the normal functions of these cells in the body. [6]

Features:

- 1 Simple/Unspecialised;
- 2 Able to divide to give new cells;
- 3 Capable of differentiating into specialised cells under appropriate conditions;

@1 mark, max 2 mk

Normal functions:

- 4 Cells from the 4-day embryo;
 - 5 Totipotent/able to give rise to a full organism;
 - 6 Embryonic stem cells;
 - 7 Derived from the inner cell mass of developing embryos/blastocysts;
 - 8 Pluripotent/able to give rise to all cell types;
 - 9 To give rise to the different cell types/tissues/organs in the developing foetus;
 - 10 Adult stem cells;
 - 11 Found in tissues/organs in the organisms;
 - 12 Non-dividing until activated by tissue injury;
 - 13 Multipotent/able to give rise to cell types found in the tissue from which they are found in;
 - 14 For maintenance and repair of tissue in which they are found;
 - 15 Examples: Bone marrow stromal (mesenchymal) cells for bone/cartilage/connective tissue;
 - 16 Hematopoietic stem cells for the different types of blood cells;
 - 17 Neural stem cells for different types of neurons;
 - 18 Epithelial stems cells for different types of specialised cells in small intestine;
 - 19 Skin stem cells for repair and replacement of the skin;
- @ ½ mark, max 4 mks

- (b) What are the benefits that can be derived from sequencing the human genome? [7]

A. *Molecular medicine (no marks for heading! Max 2 mks, @ 1 mk)*

- 1 Earlier diagnosis/detection of genetic diseases;
- 2 Gene therapy;
- 3 Rational drug design / control systems for drugs / rational drug design / pharmacogenomics 'custom drugs' ;

B. *Energy and Environmental Applications (max 1 mk, @ 1 mk)*

- 4 Use microbial genomics research to create new energy sources (biofuels);
- 5 Use microbial genomics research to develop environmental monitoring techniques to detect pollutants ;
- 6 Use microbial genomics research for safe, efficient environmental remediation;

C. *DNA Forensics (max 3 mk, @ 1 mk)*

- 7 Identify potential suspects whose DNA may match evidence left at crime scenes;
- 8 Exonerate persons wrongly accused of crimes;
- 9 Identify crime and catastrophe victims;
- 10 Establish paternity and other family relationships;
- 11 Identify endangered and protected species as an aid to wildlife officials (could be used for prosecuting poachers);

- 12 Detect bacteria and other organisms that may pollute air, water, soil, and food;**
- 13 Match organ donors with recipients in transplant programs;**
- 14 Determine pedigree for seed or livestock breeds;**
- 15 Authenticate consumables such as caviar and wine;**

D. Agriculture, Livestock Breeding, and Bioprocessing (max 1 mk, @ 1 mk)

- 16 Disease-, insect-, and drought-resistant crops;**
- 17 Healthier, more productive, disease-resistant farm animals;**
- 18 More nutritious produce ;**
- 19 Biopesticides;**
- 20 Edible vaccines incorporated into food products;**
- 21 New environmental cleanup uses for plants like tobacco;**

E. Bioarchaeology, anthropology, evolution and human migration (max 1 mk, @ 1 mk)

- 22 Human evolution through germline mutations in lineages;**
- 23 Study of migration of diff pop groups based on female genetic inheritance/lineage and migration of males via Y chromosomes;**
- 24 Compare breakpoints in the evolution of mutations with ages of populations and historical events;**

F. Risk assessment (@ 1 mk)

- 25 Assess health damage and risks caused by radiation exposure/mutagenic chemicals/cancer-causing toxins;**

Total max: 7 mk

(c) Discuss briefly the ethical issues raised by the human genome project. [7]

- 1 Fairness in the use of genetic information by insurers, employers, courts, schools, adoption agencies, and the military, among others;**
- 2 Privacy and confidentiality of genetic information;**
- 3 Possible discrimination for individuals with known genetic defects/prone to genetic diseases in different areas of life;**
- 4 Psychological impact and stigmatization due to an individual's genetic differences;**
- 5 Reproductive issues such as the adequacy of information for parents/individuals and the rights for decision making for complex and potentially controversial procedures;**
- 6 The reliability of the genetic tests for pre-natal decision making, eg. abortion.**
- 7 Uncertainties associated with gene tests for susceptibilities and complex conditions (e.g., heart disease) linked to multiple genes and gene-environment interactions.**
- 8 Conceptual and philosophical implications regard human responsibility/degree of genetic influence on behaviour and hence the party responsible/the line between medical disease and enhancement;**
- 9 Health and environmental issues concerning genetically modified foods (GM) and microbes.**
- 10 Property rights/Commercialization of products (patents, copyrights, and trade secrets)**
- 11 Fairness in the accessibility of data/material/technologies.**

@ 1mk , max 7